

CDAC MUMBAI

Concepts of Operating System

Assignment 1

- a) Navigate and List: Start by navigating to your home directory and list its contents. Then, move into a directory named "LinuxAssignment" if it exists; otherwise, create it.
- b) File Management: Inside the "LinuxAssignment" directory, create a new file named "file1.txt". Display its contents.
- c) Directory Management: Create a new directory named "docs" inside the "LinuxAssignment" directory.
- d) Copy and Move Files: Copy the "file1.txt" file into the "docs" directory and rename it to "file2.txt".
- e) Permissions and Ownership:
 - a. Change the permissions of "file2.txt" to allow read, write, and execute permissions for the owner and only read permissions for others. Then, change the owner of "file2.txt" to the current user.
- f) Final Checklist: Finally, list the contents of the "LinuxAssignment" directory and the root directory to ensure that all operations were performed correctly.
- g) File Searching:
 - a. Search for all files with the extension ".txt" in the current directory and its subdirectories.
 - b. Display lines containing a specific word in a file (provide a file name and the specific word to search).
- h) System Information:
 - a. Display the current system date and time.
- i) Networking:
 - a. Display the IP address of the system.
 - b. Ping a remote server to check connectivity (provide a remote server address to ping).
- j) File Compression:
 - a. Compress the "docs" directory into a zip file.
 - b. Extract the contents of the zip file into a new directory.

k) File Editing:

- Open the "file1.txt" file in a text editor and add some text to it.
- Replace a specific word in the "file1.txt" file with another word (provide the original word and the word to replace it with).

Code

A) Navigate and List

```
rahulpatil@rahulpatil: ~/LinuxAssignment
rahulpatil@rahulpatil:~$ ls
'Ai in eyecare.pdf'  LibrePCB-Workspace  R
cap_%d              Music               snap
Desktop             nltk_data           Templates
Docker.docx         packages-microsoft-prod.deb 'udo systemctl enable sssd-autofs.service'
Documents            Pictures            Videos
Downloads            Pro_env
genometools          Public
rahulpatil@rahulpatil:~$ if [ ! -d "LinuxAssignment" ]; then
> mkdir LinuxAssignment
> fi
rahulpatil@rahulpatil:~$ cd LinuxAssignment
```

B) - File Management

```
rahulpatil@rahulpatil:~/LinuxAssignment$ touch file1.txt
rahulpatil@rahulpatil:~/LinuxAssignment$ nano file1.txt
rahulpatil@rahulpatil:~/LinuxAssignment$ cat file1.txt
Hi
Hello Bitch
Hope you are okay and life goes wit version 26/02/2025
rahulpatil@rahulpatil:~/LinuxAssignment$
```

C)- Directory Management & Copy and Move Files

```
rahulpatil@rahulpatil:~/LinuxAssignment$ mkdir docs
rahulpatil@rahulpatil:~/LinuxAssignment$ cp file1.txt docs/file2.txt
rahulpatil@rahulpatil:~/LinuxAssignment$
```

E)- Permissions and Ownership

```
rahulpatil@rahulpatil:~/LinuxAssignment$ chmod 744 docs/file2.txt
rahulpatil@rahulpatil:~/LinuxAssignment$ sudo chown $(whoami):$(whoami) docs/file2.txt
[sudo] password for rahulpatil:
```

F) - Final Checklist & File Searching

```
rahulpatil@rahulpatil:~/LinuxAssignment$ ls
docs  file1.txt
rahulpatil@rahulpatil:~/LinuxAssignment$ find . -name "*.txt"
./file1.txt
./docs/file2.txt
```

H)- System Information & Networking

```
rahulpatil@rahulpatil:~/LinuxAssignment$ date
Wed Feb 26 10:47:01 PM IST 2025
rahulpatil@rahulpatil:~/LinuxAssignment$ ifconfig
enp2s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 10:ff:e0:5b:6f:ed txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 727 bytes 80626 (80.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 727 bytes 80626 (80.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlx3068933edf08: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.100.80 netmask 255.255.255.0 broadcast 192.168.100.255
    inet6 2402:3a80:4198:a3d0:2967:3a48:c3e:4fb4 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::8e60:dfb8:84ac:4138 prefixlen 64 scopeid 0x20<link>
    inet6 2402:3a80:4198:a3d0:7f8e:9925:6f5f:ce65 prefixlen 64 scopeid 0x0<global>
    ether 30:68:93:3e:df:08 txqueuelen 1000 (Ethernet)
    RX packets 13402 bytes 13390792 (13.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 6683 bytes 2008298 (2.0 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
rahulpatil@rahulpatil:~/LinuxAssignment$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=34.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=35.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=32.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=31.4 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 31.350/33.440/35.474/1.617 ms
```

J) File Compression & File Editing:

```
rahulpatil@rahulpatil:~/LinuxAssignment$ zip -r docs.zip docs
  adding: docs/ (stored 0%)
  adding: docs/file2.txt (stored 0%)
rahulpatil@rahulpatil:~/LinuxAssignment$ mkdir extracted_docs
unzip docs.zip -d extracted_docs
Archive:  docs.zip
  creating: extracted_docs/docs/
  extracting: extracted_docs/docs/file2.txt
rahulpatil@rahulpatil:~/LinuxAssignment$ nano file1.txt
rahulpatil@rahulpatil:~/LinuxAssignment$ sed -i 's/oldword/newword/g' file1.txt
rahulpatil@rahulpatil:~/LinuxAssignment$
```

a. Suppose you have a file named "data.txt" containing important information. Display the first 10 lines of this file to quickly glance at its contents using a command.

Code: `head -n 10 data.txt`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ head -n 10 data.txt
Afghanistan
Australia
Brazil
Canada
China
Denmark
Egypt
France
Germany
India
```

b. Now, to check the end of the file for any recent additions, display the last 5 lines of "data.txt" using another command.

Code : `tail -n 5 data.txt`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ tail -n 5 data.txt
South Korea
Spain
United Kingdom
United States
Venezuela
```

c. In a file named "numbers.txt," there are a series of numbers. Display the first 15 lines of this file to analyze the initial data set.

Code: `head -n 15 numbers.txt`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ head -n 15 numbers.txt
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
```

d. To focus on the last few numbers of the dataset, display the last 3 lines of "numbers.txt".

Code: `tail -n 3 numbers.txt`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ tail -n 3 numbers.txt
434946534963459
3456374963459435649
```

e. Imagine you have a file named "input.txt" with text content. Use a command to translate all lowercase letters to uppercase in "input.txt" and save the modified text in a new file named "output.txt."

Code: `tr 'a-z' 'A-Z' < input.txt > output.txt`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ tr 'a-z' 'A-Z' < input.txt > output.txt
rahulpatil@rahulpatil:~/Concept of OS Assignment$ ls
Assingment1.docx  data.txt  docs.zip      extracted_docs  fruit.txt  numbers.txt
Assingment1.pdf  docs      duplicate.txt  file1.txt      input.txt  output.txt
rahulpatil@rahulpatil:~/Concept of OS Assignment$ cat output.txt
HELLO, THIS IS A SAMPLE TEXT FILE.
THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG.
HAVE A GREAT DAY!
```

f. In a file named "duplicate.txt," there are several lines of text, some of which are duplicates. Use a command to display only the unique lines from "duplicate.txt."

Code: `sort duplicate.txt | uniq`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ sort duplicate.txt | uniq
apple
banana
grape
orange
pear
```

g. In a file named "fruit.txt," there is a list of fruits, but some fruits are repeated. Use a command to display each unique fruit along with the count of its occurrences in "fruit.txt."

Code `sort fruit.txt | uniq -c`

```
rahulpatil@rahulpatil:~/Concept of OS Assignment$ sort fruit.txt | uniq -c
  3 apple
  2 banana
  2 grape
  1 orange
  1 pear
```